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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Com (BM) DEGREE EXAMINATION APRIL 2021

ASCM301S - BUSINESS STATISTICS

Time : 3 hrs **Max Marks :75**

SECTION - A (10x2=20)

Answer ALL the questions

1. Define primary data.

2. Name the types of classification.

3. For the set of observations 13, 25, 36, 22, 18, 45, 21, 26,30, 22, find range and coefficient of range.

4. The formula for mean deviation is _____.

5. Define Correlation.

6. What are the two Regression equations?

7. Define index numbers.

8. What are the components of Time series?

9. Define chance causes of variation.

10. Give the control limits for R-chart.

SECTION - B (5 x 5 =25)

Answer **ALL** the questions

11. (a) The following data gives the number of children in 50 families. Construct a suitable frequency table.

4	2	0	2	3	2	2	1	0	2
3	5	1	1	4	2	1	3	4	2
6	1	2	2	2	1	3	4	1	0
2	4	0	0	1	3	6	1	0	1
1	3	1`	1	0	1	2	2	2	5

(Or)

(b) Explain collection of Primary data.

12. (a) For a distribution Bowley's coefficient of skewness is -0.36, lower quartile is 8.6 and median is 12.3. What is its quartile co-efficient of dispersion?

(Or)

(b) Find Standard Deviation for the following data.

X : 10 20 30 40 50

Y: 2 8 10 7 3

13. (a) Calculate coefficient of correlation by the method of concurrent deviation from the following data:

X : 84 85 62 48 84 95 103 100 85 115

Y : 20 23 19 21 25 25 28 26 26 30

(Or)

(b)The following table gives age(X) in years of cars and annual maintenance cost (Y) (in hundred rupees).

X ; 1 3 5 7 9

Y : 15 18 21 23 22

Estimate the maintenance cost for a 4-year old car after finding the regression equations..

14. (a) In the construction of certain cost of living index number, the following group index number were found. Calculate the cost of living index number by using weighted arithmetic geometric mean.

Group	Index Number	Weights
Food	352	48
Clothing	200	10
House Rent	230	8
Fuel and Lighting	160	12
Miscellaneous	190	15

(Or)

(b) Calculate the seasonal indices from the following data using the average method.

Year	I Quarter	II Quarter	III Quarter	IV Quarter
2001	72	68	80	70
2002	76	70	82	74
2003	74	66	84	80
2004	76	74	84	78
2005	78	74	86	82

15. (a) A machine is set to deliver the packets of a given weight. Ten samples of size five each were examined and the following results were obtained:

Sample No. : 1 2 3 4 5 6 7 8 9 10

Mean : 43 49 37 44 45 37 51 46 43 47

Calculate the values for the central line and the control limits for the mean chart. Comment on the state of control.

(Or)

(b) Give any five applications of c-chart.

SECTION - C (3 X 10 = 30)

Answer **any THREE** questions

16. Find the Mean,Median and Mode for the following data and verify the empirical relation.

Class	1-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
Frequency	3	7	13	17	12	10	8	8	6	6

17. The scores of two players A and B in 12 rounds are given below:

A: 74 75 78 72 78 77 79 81 79 76 72 71

B: 87 84 80 88 89 85 86 82 82 79 86 80

Identify the better player and the more consistent player.

18. A random sample of recent repair jobs was selected and estimated cost and actual cost were recorded.

Estimated cost : 300 450 800 250 500 975 475 400

Actual cost : 273 486 734 297 631 872 396 457

Calculate the value of Spearman's Rank Correlation Coefficient.

19. Construct index number of prices from the following data using (a) Laspeyre's and (b) Fisher's methods.

Commodity	2000		2010	
	Price (in Rs.)	Quantity	Price (in Rs.)	Quantity
A	6	50	10	56
B	2	100	2	120
C	4	60	6	60
D	10	30	12	24
E	8	40	12	36

20. 20 samples each of size 10 were inspected. The number of defectives detected in each of them is given below:

Sample No.	1	2	3	4	5	6	7	8	9	10
No.of defectives	0	1	0	3	9	2	0	7	0	1
Sample No.	11	12	13	14	15	16	17	18	19	20
No. of defectives	1	0	0	3	1	0	0	2	1	0

Construct the number of defectives chart.

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